Post-Quantum SUCI Migration for 5G/6G E-Business Services*

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Abstract

Subscriber identity protection is a critical requirement for 5G and future 6G e-business services such as mobile payments and digital identity wallets. In 5G systems, the Subscription Permanent Identifier (SUPI) is concealed as a Subscription Concealed Identifier (SUCI) before transmission. However, current implementations rely on classical elliptic-curve public-key cryptography, which is vulnerable to quantum attacks. Recent 3GPP proposals have explored several Post-Quantum Cryptography (PQC) migration approaches, including Standalone PQC, Hybrid PQC, Hybrid Nested PQC, Symmetric Cryptography, and Quantum Channel. In this study, we categorize these five migration paths into a single comparison space and evaluate their practical deployment trade-offs, identifying key architectural bottlenecks for future e-business environments.

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